IRONKERNEL EMBEDDED

NUCLEO-F401RE Demo Brief
Revision 11-13-2026

VITA 46.11 IPMC in Rust

Try the public Tier-1 IPMC on an STM32 NUCLEO-F401RE. Free evaluation, a no-fee Open Product option, and paid commercial licenses.

The IronKernel IPMC is management-controller firmware for VPX plug-in modules. The public IPMIRustFRU repository supplies the Tier-1 firmware, bootloader, board support, PC emulator and host tools. This demonstration targets the STM32F401RE Arm Cortex-M4F on the NUCLEO-F401RE board. Firmware is written in Rust, with no heap, no RTOS and no C runtime. The protocol logic forbids unsafe Rust; register access belongs to the board support.

Tier-1 answers the Chassis Manager, reports identity, FRU inventory and sensor readings, and maintains the Tier-1 FRU state. The private Tier-2/3 extension adds event reporting and further management functions under a separate paid agreement. The public demo contains Tier-1 only; paid Tier-2/3 customers receive the full source under a commercial license.

What the Tier-1 demonstration includes

Area	Demonstration
Identity and inventory	Device ID, self test, FRU inventory read/write and geographic address information
Sensors	VITA 46.11 Tier-1 sensors, SDR records, supply-rail readings and die temperature
IPMB	IPMB-A on the standard bench setup; dual-bus support configurable in the source
Persistent FRU	Board and Product areas, flash-backed writes and write-protection handling
Boot and update	Public Tier-1 bootloader, image length and SHA-256 validation, firmware update over the console CLI or application UART
Diagnostics	Console, application UART, IPMB statistics and capture tools
PC evaluation	The same protocol logic in the emulator, without the NUCLEO hardware

The default bench configuration uses IPMB-A. IPMB-B hardware testing needs the second bus wired and enabled. Rail readings require the documented divider wiring or a deliberately configured simulated source. The NUCLEO board is a development platform; a production VPX module needs its own board support, wiring and integration checks.

Protocol tiers at a glance

Package	Coverage	Availability
Tier-1	Discovery, inventory, sensor readings and Tier-1 FRU management	Public source and NUCLEO demo; terms on page 3
Tier-2	Tier-1 plus event receiver configuration, outbound Platform Event messages, Cold Reset and FRU Control Capabilities	Included in the combined Tier-2/3 commercial package
Tier-3	Tier-1 and Tier-2 plus activation and policy, payload modes, event controls, SEL and further Tier-3 commands	Included in the combined Tier-2/3 commercial package

NUCLEO-F401RE setup and image loading

Use a **NUCLEO-F401RE**, USB data cable and serial terminal. ST-LINK exposes the programming drive and console COM port. Physical IPMI requests need a Chassis Manager or IPMB adapter; the application UART needs a separate 3.3 V USB-to-UART adapter.

Signal	NUCLEO pin	Connection
IPMB-A SCL	PB8, D15	Chassis Manager / IPMB adapter SCL
IPMB-A SDA	PB9, D14	Chassis Manager / IPMB adapter SDA
Console	PA2 TX / PA3 RX	Onboard ST-LINK virtual COM port, 115200 baud, 8N1
Application UART	PB6 TX / PB7 RX	Separate adapter RX / TX and common ground
Ground	GND	Common ground with the IPMB or UART adapter

Use open-drain I2C wiring and suitable pull-ups. Keep Aardvark target power off and connect only SCL, SDA and GND for this IPMB-A test. Do not apply payload rail voltage directly to an ADC pin: A0 needs a 4:1 divider and A1/A2/A3 need 2:1 dividers, with ADC inputs kept within the MCU supply range. External rail wiring is separate from the basic USB-and-IPMB demonstration.

Image files

The packaging command produces two checked Tier-1 images, hashes and a build manifest in `images/`. See image availability for this checkout.

File	Use
`images/ipmc-board-tier1.bin`	Initial installation through the ST-LINK programming drive; bootloader and application combined
`images/ipmc-nucleo-f401re-tier1.hex`	Subsequent application update using the bootloader and `fwupdate.py`
`images/SHA256SUMS.txt`	SHA-256 checksums for both images
`images/manifest.json`	Build date, source revision, dirty-worktree state, toolchain and Tier

1. Check each image's SHA-256 digest against `SHA256SUMS.txt`.
2. Connect ST-LINK USB and find its programming drive, normally `NOD_F401RE`.
3. Copy `ipmc-board-tier1.bin` to the drive. It installs bootloader and runtime at `0x08000000` and **erases saved FRU inventory and settings**.
4. Wait for programming and reset; check `FAIL.TXT` if it fails. Open the console COM port at 115200 baud, 8N1; try `HELP`, `IPMB STATS`, `SENSORS` and `FRU SHOW`.

Do not copy the application HEX file or an application-only binary to the ST-LINK drive. The HEX file is for the UART update tool; its application address is `0x0800C000`. A firmware hash verifies integrity, not publisher identity or a digital signature.

Firmware update over the CLI: included in public Tier-1

The public Tier-1 release includes firmware update through the console CLI as well as the application UART. From the runtime console, `DEBUG BOOTLOADER` enters the bootloader; when interrupting payload power requires confirmation, use `DEBUG BOOTLOADER CONFIRM`. The public host tool automates entry, application erase, Intel HEX loading, SHA-256 verification and restart. From the repository root, replacing COM3 with the console port:

```
python tools/fwupdate/fwupdate.py --port COM3 --uart cli `
  --enter cli --enter-port COM3 update `
  demos/nucleo-f401re/images/ipmc-nucleo-f401re-tier1.hex
```

An ordinary application update preserves the bootloader and stored FRU data; restarting into the bootloader interrupts the application and can drop payload power. This feature needs no Tier-2/3 extension. It is a CLI/bootloader facility, distinct from standardized IPMI/VSO commands in

the command-support appendix.

License choices and fees

Two paid offerings are available: **Tier-1** and **Tier-2/3 (Tier-1 included)**. Tier-2 is not sold separately. All prices below are **USD**, per product/SKU. Paid license fees are one-time; there are **no royalties and no per-unit fees**. The signed commercial agreement identifies the licensed product/SKU, deliverables and deployment rights. A different product/SKU needs its own license unless an enterprise agreement expressly covers it. **Enterprise licensing for multiple product SKUs is available upon request and quotation.**

Option	License fee	Annual maintenance	What it permits
Tier-1 evaluation	**Free**	**Free public updates**	Evaluation and development, unsold prototypes, research, teaching, hobby and personal use under section 2 of the public license
Tier-1 Open Product	**Free**	**Free public updates**	Product deployment while every public-source, contribution and notice condition of section 3 is met
Tier-1 commercial	**$5,000**	**Free**	Deployment under a paid agreement when the Open Product conditions are not met, including a product with private Design Sources
Tier-2/3 commercial (Tier-1 included)	**$25,000**	**$5,000/year**	Full Tier-1/2/3 deployment under a paid agreement; full source included; private commercial license

Tier-1 has no annual maintenance charge: updates are available from the public repository. This does not promise a support response time or custom engineering. Tier-2/3 maintenance covers updates and support as specified in the agreement; the agreement sets the start date, coverage and renewal terms. Board ports, custom sensors, FRU/SDR work and integration services are separately quoted. A paid Tier-1 license is credited in full toward a Tier-2/3 license for the same product/SKU.

The no-fee Tier-1 Open Product conditions

The public repository uses the **IPMIRustFRU Source-Available License**. Commercial use can qualify for the no-fee Open Product option, but access to the source alone is not enough:

1. Publish the Product's **complete Design Sources**, including your own parts of the design, at no charge and without registration under the public license, except components that retain their third-party licenses. Publish them no later than the first shipment, lease, licensing, customer deployment or paid-service use, and publish every later release too.
2. Offer every Change to IPMIRustFRU back to the IronKernel repository as a pull request within **90 days** of the Product release containing it. Acceptance is not required.
3. Name IPMIRustFRU in the Product documentation and link both this repository and the published Design Sources. Keep the required notices and mark modified files with a date.

These conditions continue while the Product is supplied or used to provide a paid service. If you need to keep the Product's Design Sources private, or cannot meet every condition, obtain a paid agreement before commercial delivery or deployment. The public license also restricts offering IPMIRustFRU or a work mainly based on it as a competing IPMC software product. A paid agreement must expressly cover any different rights you need.

This is a source-available license with commercial conditions, rather than an OSI open source license. The public LICENSE.md is the governing text for Tier-1. A signed commercial agreement governs the rights it grants and any differences from that text. Contact **ironkernel.com** for commercial licenses, maintenance and engineering services.

Paid extension, build and verification

Tier-2 and Tier-3 stay private

The Tier-2/3 repository stays private. **Paid Tier-2/3 customers receive the full source code** for their licensed package under a commercial agreement. The license permits building, modifying and using that source for the licensed product/SKU, or the products/SKUs covered by an enterprise agreement. Copyright and ownership remain with IronKernel and the respective rights holders.

Delivered source and derived source files remain confidential. Access is limited to employees and contractors who need it for licensed work and are bound by confidentiality and use restrictions. Customers may deliver the agreed compiled firmware with the licensed Product, but may not publish or resell the source or sublicense the extension to others. Neither the Tier-1 evaluation option nor the Open Product option licenses the extension.

This public demo is Tier-1. It contains no Tier-2/3 firmware images. Any commercial Tier-2/3 image delivery requires the separate agreement; the public demo license grants no right to unlock or use paid features. Licensed standards documents and private reference material are also outside the public license.

Build the Tier-1 demo package

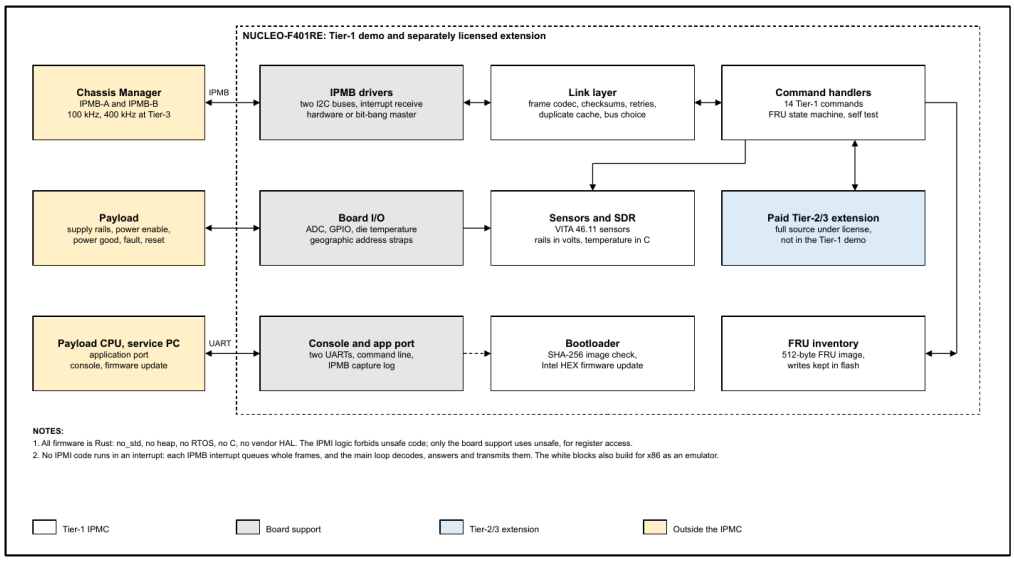
From the repository root in PowerShell:

```
powershell -NoProfile -ExecutionPolicy Bypass -File demos/nucleo-f401re/package.ps1
```

Install the Rust toolchain pinned in `rust-toolchain.toml`, the `thumbv7em-none-eabihf` target and GNU Arm binutils (`arm-none-eabi-objcopy` and `arm-none-eabi-size`) first. The packager sets `IPMC_TIER23=0` for the build, so the private extension is excluded even on a developer PC where it is present. It restores the previous environment afterward. It builds and checks the images, checks Tier-1 identity, and copies only the combined board binary and application HEX file into this demo. It does not flash a connected board.

For firmware design details and independent checks, see the software specification, command list, test-coverage report and Chassis Manager tools. Historical test results describe their recorded builds; they are not a test certificate for newly rebuilt demo images. On-board dual-bus checks need the corresponding hardware setup.

Architecture



The IPMI logic also builds for a PC emulator. A port to another MCU replaces the board support and must be validated for that board; this demo makes no portability or certification claim for an untested target. Ferrocene qualification and DO-178C acceptance are not claimed for these demo images.

Complete VITA 46.11-2022 command support

The catalog follows the IPMI and VSO groups of VITA 46.11-2022 Tables 10.1.1-1 and 10.1.2-1. Unsupported, optional, reserved and Chassis-Manager-only entries remain visible. This product is an IPMC, not a Chassis Manager.

Req T1/T2/T3: M = mandatory; O = optional; NA = not applicable; O/M = conditional mandatory support. Bracketed numbers refer to the notes after the tables. Requirements and implementation are separate columns.

Impl T1/T2/T3: Yes = received command implemented; No = not implemented; Transmit only = outbound operation, with incoming requests unsupported. Unimplemented addressed commands normally return C1h. Reserved entries are not commands. Conditional requirements depend on the interfaces and FRUs fitted.

Implemented received command forms: **14 at Tier-1, 18 at Tier-2, 36 at Tier-3**, counting Broadcast Get Device ID separately from unicast. Outbound Platform Event generation is additional at Tier-2/3. The paid Tier-2/3 package includes both configurations; Tier-2 is not sold separately.

Mandatory-command coverage: no entry marked M or M[n] is marked No. Conditionally mandatory O/M entries must be assessed against the scope below; this table is an implementation statement, not a conformance certificate.

IPMI references retain the source table's version and section notation. VSO references identify VITA 46.11-2022 sections; VSO requests use NetFn 2Ch and group identifier 03h. See the public command document for further context.

IPMI commands: VITA Table 10.1.1-1

IPM Device "Global" Commands

NetFn/Cmd	Command	Reference	Req T1	Req T2	Req T3	Impl T1	Impl T2	Impl T3
06h/00h	Reserved	NA	NA	NA	N/A	No	No	No
06h/01h	Get Device ID	1.5:17.1	M	M	M	Yes	Yes	Yes
06h/02h	Cold Reset	1.5:17.2	O[2]	M[2]	M	No	Yes	Yes
06h/03h	Warm Reset	1.5:17.3	O	O	M	No	No	Yes
06h/04h	Get Self Test Results	1.5:17.4	M	M	M	Yes	Yes	Yes
06h/05h	Manufacturing Test On	1.5:17.5	O	O	O	No	No	No
06h/06h	Set ACPI Power State	1.5:17.6	O	O	O	No	No	No
06h/07h	Get ACPI Power State	1.5:17.7	O[2]	O[2]	O	No	No	No
06h/08h	Get Device GUID	1.5:17.8	O	O	O	No	No	No
06h/01h	Broadcast "Get Device ID"[3]	1.5:17.9	M	M	M	Yes	Yes	Yes

BMC Watchdog Timer Commands

NetFn/Cmd	Command	Reference	Req T1	Req T2	Req T3	Impl T1	Impl T2	Impl T3
06h/22h	Reset Watchdog Timer	1.5:21.5	O	O	O	No	No	No
06h/24h	Set Watchdog Timer	1.5:21.6	O	O	O	No	No	No
06h/25h	Get Watchdog Timer	1.5:21.7	O	O	O	No	No	No

BMC Device and Messaging Commands

NetFn/Cmd	Command	Reference	Req T1	Req T2	Req T3	Impl T1	Impl T2	Impl T3
06h/2Eh	Set BMC Global Enables[6]	1.5:18.1	O	O	O/M[7]	No	No	No
06h/2Fh	Get BMC Global Enables	1.5:18.2	O	O	M	No	No	Yes
06h/30h	Clear Message Flags[6]	1.5:18.3	O	O	O	No	No	No
06h/31h	Get Message Flags	1.5:18.4	O	O	O	No	No	No
06h/32h	Enable Message Channel Receive[6]	1.5:18.5	O	O	O	No	No	No
06h/33h	Get Message[6]	1.5:18.6	O	O	O	No	No	No
06h/34h	Send Message	1.5:18.7	O	O	O/M[7]	No	No	No
06h/35h	Read Event Message Buffer[6]	1.5:18.8	O	O	O	No	No	No
06h/36h	Get BT Interface Capabilities	1.5:18.9	O[2]	O[2]	O[2]	No	No	No
06h/52h	Master Write-Read	1.5:18.10	O[2]	O[2]	O/M[8]	No	No	No
06h/37h	Get System GUID	1.5:18.13	O[2]	O[2]	O[2]	No	No	No
06h/38h	Get Channel Authentication Capabilities	1.5:18.12	O[2]	O[2]	O[2]	No	No	No
06h/39h	Get Session Challenge	1.5:18.14	O[2]	O[2]	O[2]	No	No	No
06h/3Ah	Activate Session	1.5:18.15	O[2]	O[2]	O[2]	No	No	No
06h/3Bh	Set Session Privilege Level	1.5:18.16	O[2]	O[2]	O[2]	No	No	No
06h/3Ch	Close Session	1.5:18.17	O[2]	O[2]	O[2]	No	No	No
06h/3Dh	Get Session Info	1.5:18.18	O[2]	O[2]	O[2]	No	No	No
06h/3Fh	Get AuthCode	1.5:18.19	O	O	O	No	No	No
06h/40h	Set Channel Access	1.5:18.20	O[2]	O[2]	O[2]	No	No	No
06h/41h	Get Channel Access	1.5:18.21	O[2]	O[2]	O[2]	No	No	No
06h/42h	Get Channel Info	1.5:18.18	O[2]	O[2]	O[2]	No	No	No
06h/43h	Set User Access	1.5:18.23	O[2]	O[2]	O[2]	No	No	No
06h/44h	Get User Access	1.5:18.24	O[2]	O[2]	O[2]	No	No	No
06h/45h	Set User Name	1.5:18.25	O[2]	O[2]	O[2]	No	No	No
06h/46h	Get User Name	1.5:18.26	O[2]	O[2]	O[2]	No	No	No
06h/47h	Set User Password	1.5:18.27	O[2]	O[2]	O[2]	No	No	No

Chassis Device Commands

NetFn/Cmd	Command	Reference	Req T1	Req T2	Req T3	Impl T1	Impl T2	Impl T3
00h/00h	Get Chassis Capabilities	1.5:22.1	O	O	O	No	No	No
00h/01h	Get Chassis Status	1.5:22.2	O	O	O	No	No	No
00h/02h	Chassis Control	1.5:22.3	O	O	O	No	No	No
00h/03h	Chassis Reset	1.5:22.4	O	O	O	No	No	No
00h/04h	Chassis Identify	1.5:22.5	O	O	O	No	No	No
00h/05h	Set Chassis Capabilities	1.5:22.6	O	O	O	No	No	No
00h/06h	Set Power Restore Policy	1.5:22.7	O	O	O	No	No	No
00h/07h	Get System Reset Cause	1.5:22.9	O[2]	O[2]	O[2]	No	No	No

NetFn/Cmd	Command	Reference	Req T1	Req T2	Req T3	Impl T1	Impl T2	Impl T3
00h/08h	Set System Boot Options	1.5:22.10	O[2]	O[2]	O[2]	No	No	No
00h/09h	Get System Boot Options	1.5:22.11	O[2]	O[2]	O[2]	No	No	No
00h/0Fh	Get POH Counter	1.5:22.12	O	O	O	No	No	No

Event Commands

NetFn/Cmd	Command	Reference	Req T1	Req T2	Req T3	Impl T1	Impl T2	Impl T3
04h/00h	Set Event Receiver	1.5:23.1	NA	M	M	No	Yes	Yes
04h/01h	Get Event Receiver	1.5:23.2	NA	M	M	No	Yes	Yes
04h/02h	Platform Event (a.k.a "Event Message")	1.5:23.3	NA	M	M	No	Transmit only	Transmit only

PEF and Alerting Commands

NetFn/Cmd	Command	Reference	Req T1	Req T2	Req T3	Impl T1	Impl T2	Impl T3
04h/10h	Get PEF Capabilities	1.5:24.1	NA	O[2]	O[2]	No	No	No
04h/11h	Arm PEF Postpone Timer	1.5:24.2	NA	O[2]	O[2]	No	No	No
04h/12h	Set PEF Configuration Parameters	1.5:24.3	NA	O[2]	O[2]	No	No	No
04h/13h	Get PEF Configuration Parameters	1.5:24.4	NA	O[2]	O[2]	No	No	No
04h/14h	Set Last Processed Event ID	1.5:24.5	NA	O[2]	O[2]	No	No	No
04h/15h	Get Last Processed Event ID	1.5:24.6	NA	O[2]	O[2]	No	No	No
04h/16h	Alert Immediate	1.5:24.7	NA	O[2]	O[2]	No	No	No
04h/17h	PET Acknowledge	1.5:24.8	NA	O[2]	O[2]	No	No	No

Sensor Device Commands

NetFn/Cmd	Command	Reference	Req T1	Req T2	Req T3	Impl T1	Impl T2	Impl T3
04h/20h	Get Device SDR Info	1.5:29.2	M	M	M	Yes	Yes	Yes
04h/21h	Get Device SDR	1.5:29.3	M[2]	M[2]	M[2]	Yes	Yes	Yes
04h/22h	Reserve Device SDR Repository	1.5:29.4	M[2]	M[2]	M[2]	Yes	Yes	Yes
04h/23h	Get Sensor Reading Factors	1.5:29.5	O[2]	O[2]	O[2]	No	No	No
04h/24h	Set Sensor Hysteresis	1.5:29.6	O	O	O	No	No	No
04h/25h	Get Sensor Hysteresis	1.5:29.7	O	O	O	No	No	No
04h/26h	Set Sensor Threshold	1.5:29.8	O	O	O	No	No	No
04h/27h	Get Sensor Threshold	1.5:29.9	O[2]	O[2]	O[2]	No	No	No
04h/28h	Set Sensor Event Enable	1.5:29.10	NA	O	M	No	No	Yes
04h/29h	Get Sensor Event Enable	1.5:29.11	NA	O[2]	M	No	No	Yes
04h/2Ah	Re-arm Sensor Events	1.5:29.12	NA	O[2]	O[2]	No	No	No
04h/2Bh	Get Sensor Event Status	1.5:29.13	NA	O	O	No	No	No
04h/2Dh	Get Sensor Reading	1.5:29.14	M	M	M	Yes	Yes	Yes

NetFn/Cmd	Command	Reference	Req T1	Req T2	Req T3	Impl T1	Impl T2	Impl T3
04h/2Eh	Set Sensor Type	1.5:29.15	O	O	O	No	No	No
04h/2Fh	Get Sensor Type	1.5:29.16	O[2]	O[2]	O[2]	No	No	No
04h/30h	Set Sensor Reading	2.0:35.17	O	O	O	No	No	No

FRU Device Commands

NetFn/Cmd	Command	Reference	Req T1	Req T2	Req T3	Impl T1	Impl T2	Impl T3
0Ah/10h	Get FRU Inventory Area Info	1.5:28.1	M	M	M	Yes	Yes	Yes
0Ah/11h	Read FRU Data	1.5:28.2	M	M	M	Yes	Yes	Yes
0Ah/12h	Write FRU Data	1.5:28.3	M	M	M	Yes	Yes	Yes

SDR Device Commands

NetFn/Cmd	Command	Reference	Req T1	Req T2	Req T3	Impl T1	Impl T2	Impl T3
0Ah/20h	Get SDR Repository Info	1.5:27.9	O	O	O	No	No	No
0Ah/21h	Get SDR Repository Allocation Info	1.5:27.10	O	O	O	No	No	No
0Ah/22h	Reserve SDR Repository	1.5:27.11	O	O	O	No	No	No
0Ah/23h	Get SDR	1.5:27.12	O[2]	O[2]	O[2]	No	No	No
0Ah/24h	Add SDR	1.5:27.13	O[2]	O[2]	O[2]	No	No	No
0Ah/25h	Partial Add SDR	1.5:27.14	O[2]	O[2]	O[2]	No	No	No
0Ah/26h	Delete SDR	1.5:27.15	O[2]	O[2]	O[2]	No	No	No
0Ah/27h	Clear SDR Repository	1.5:27.16	O[2]	O[2]	O[2]	No	No	No
0Ah/28h	Get SDR Repository Time	1.5:27.17	O[2]	O[2]	O[2]	No	No	No
0Ah/29h	Set SDR Repository Time	1.5:27.18	O[2]	O[2]	O[2]	No	No	No
0Ah/2Ah	Enter SDR Repository Update Mode	1.5:27.19	O[2]	O[2]	O[2]	No	No	No
0Ah/2Bh	Exit SDR Repository Update Mode	1.5:27.20	O[2]	O[2]	O[2]	No	No	No
0Ah/2Ch	Run Initialization Agent	1.5:27.21	O[2]	O[2]	O[2]	No	No	No

SEL Device Commands

NetFn/Cmd	Command	Reference	Req T1	Req T2	Req T3	Impl T1	Impl T2	Impl T3
0Ah/40h	Get SEL Info	1.5:25.2	O	O	M	No	No	Yes
0Ah/41h	Get SEL Allocation Info	1.5:25.3	O	O	O	No	No	No
0Ah/42h	Reserve SEL	1.5:25.4	O[2]	O[2]	M	No	No	Yes
0Ah/43h	Get SEL Entry	1.5:25.5	O	O	M	No	No	Yes
0Ah/44h	Add SEL Entry	1.5:25.6	O[2]	O[2]	O[2]	No	No	No
0Ah/45h	Partial Add SEL Entry	1.5:25.7	O[2]	O[2]	O[2]	No	No	No
0Ah/46h	Delete SEL Entry	1.5:25.8	O	O	O	No	No	No
0Ah/47h	Clear SEL	1.5:25.9	O	O	M	No	No	Yes

NetFn/Cmd	Command	Reference	Req T1	Req T2	Req T3	Impl T1	Impl T2	Impl T3
0Ah/48h	Get SEL Time	1.5:25.10	O	O	M	No	No	Yes
0Ah/49h	Set SEL Time	1.5:25.11	O	O	M	No	No	Yes
0Ah/5Ah	Get Auxiliary Log Status	1.5:25.12	O	O	O	No	No	No
0Ah/5Bh	Set Auxiliary Log Status	1.5:25.13	O[2]	O[2]	O[2]	No	No	No

LAN Device Commands

NetFn/Cmd	Command	Reference	Req T1	Req T2	Req T3	Impl T1	Impl T2	Impl T3
0Ch/01h	Set LAN Configuration Parameters	1.5:19.1	O[2]	O[2]	O[2]	No	No	No
0Ch/02h	Get LAN Configuration Parameters	1.5:19.2	O[2]	O[2]	O[2]	No	No	No
0Ch/03h	Suspend BMC ARPs	1.5:19.3	O[2]	O[2]	O[2]	No	No	No
0Ch/04h	Get IP/UDP/RMCP Statistics	1.5:19.4	O	O	O	No	No	No

Serial/Modem Device Commands

NetFn/Cmd	Command	Reference	Req T1	Req T2	Req T3	Impl T1	Impl T2	Impl T3
0Ch/10h	Set Serial/Modem Configuration	1.5:20.1	O[2]	O[2]	O[2]	No	No	No
0Ch/11h	Get Serial Modem Configuration	1.5:20.2	O[2]	O[2]	O[2]	No	No	No
0Ch/12h	Set Serial/Modem Mux	1.5:20.3	O[2]	O[2]	O[2]	No	No	No
0Ch/13h	Get TAP Response Codes	1.5:20.4	O[2]	O[2]	O[2]	No	No	No
0Ch/14h	Set PPP UDP Proxy Transmit Data	1.5:20.5	O[2]	O[2]	O[2]	No	No	No
0Ch/15h	Get PPP UDP Proxy Transmit Data	1.5:20.6	O[2]	O[2]	O[2]	No	No	No
0Ch/16h	Send PPP UDP Proxy Packet	1.5:20.7	O[2]	O[2]	O[2]	No	No	No
0Ch/17h	Get PPP UDP Proxy Packet	1.5:20.8	O[2]	O[2]	O[2]	No	No	No
0Ch/18h	Serial/Modem Connection Active	1.5:20.9	O[2]	O[2]	O[2]	No	No	No
0Ch/19h	Callback	1.5:20.10	O	O	O	No	No	No
0Ch/1Ah	Set User Callback Options	1.5:20.11	O[2]	O[2]	O[2]	No	No	No
0Ch/1Bh	Get User Callback Options	1.5:20.12	O[2]	O[2]	O[2]	No	No	No

Firmware Firewall Commands

NetFn/Cmd	Command	Reference	Req T1	Req T2	Req T3	Impl T1	Impl T2	Impl T3
06h/09h	Get NetFn Support	2.0:21.2	O[2]	O[2]	O[2]	No	No	No
06h/64h	Get OEM NetFn IANA Support	2.0:21.11	O[2]	O[2]	O[2]	No	No	No
06h/0Ah	Get Command Support	2.0:21.3	O[2]	O[2]	O[2]	No	No	No
06h/0Bh	Get Command Sub-function Support	2.0:21.4	O[2]	O[2]	O[2]	No	No	No
06h/0Ch	Get Configurable Commands	2.0:21.5	O[2]	O[2]	O[2]	No	No	No
06h/0Dh	Get Configurable Command Sub-Functions	2.0:21.6	O[2]	O[2]	O[2]	No	No	No
06h/61h	Get Command Enables	2.0:21.8	O[2]	O[2]	O[2]	No	No	No

NetFn/Cmd	Command	Reference	Req T1	Req T2	Req T3	Impl T1	Impl T2	Impl T3
06h/63h	Get Configurable Command Sub-function Enables	2.0:21.10	O[2]	O[2]	O[2]	No	No	No
06h/60h	Set Command Enables	2.0:21.7	O[2]	O[2]	O[2]	No	No	No
06h/62h	Set Configurable Command Sub-function Enables	2.0:21.9	O[2]	O[2]	O[2]	No	No	No

VSO commands: VITA Table 10.1.2-1

NetFn/Cmd	Command	Reference	Req T1	Req T2	Req T3	Impl T1	Impl T2	Impl T3
2Ch/00h	Get VSO Capabilities	§10.1.3.1	M	M	M	Yes	Yes	Yes
2Ch/01h	Get Chassis Address Table Info	§10.1.3.2	NA	NA	NA	No	No	No
2Ch/02h	Get Chassis Identifier	§10.1.3.4	NA	NA	NA	No	No	No
2Ch/03h	Set Chassis Identifier	§10.1.3.5	NA	NA	NA	No	No	No
2Ch/04h	FRU Control	§10.1.3.6	O	O	O/M[6]	No	No	Yes
2Ch/05h	Get FRU LED Properties	§10.1.3.27	O	O	O	No	No	No
2Ch/06h	Get LED Color Capabilities	§10.1.3.28	O	O	O	No	No	No
2Ch/07h	Set FRU LED State	§10.1.3.30	O	O	O	No	No	No
2Ch/08h	Get FRU LED State	§10.1.3.29	O	O	O	No	No	No
2Ch/09h	Set IPMB State	§10.1.3.7	M	M	M	Yes	Yes	Yes
2Ch/0Ah	Set FRU State Policy Bits	§10.1.3.9	O/M[2]	O/M[2]	M	No	No	Yes
2Ch/0Bh	Get FRU State Policy Bits	§10.1.3.8	O/M[2]	O/M[2]	M	No	No	Yes
2Ch/0Ch	Set FRU Activation	§10.1.3.10	O	O	M	No	No	Yes
2Ch/0Dh	Get Device Locator Record ID	§10.1.3.11	M	M	M	Yes	Yes	Yes
2Ch/0Eh-13h	(reserved)	Reserved	-	-	-	No	No	No
2Ch/14h	Get Fan Speed Properties	§10.1.3.12	O	O	O	No	No	No
2Ch/15h	Set Fan Level	§10.1.3.14	O	O	O	No	No	No
2Ch/16h	Get Fan Level	§10.1.3.13	O	O	O	No	No	No
2Ch/17h	(reserved)	Reserved	-	-	-	No	No	No
2Ch/18h	Get IPMB Link Info	§10.1.3.17	O	O	O	No	No	No
2Ch/19h-1Ah	(reserved)	Reserved	-	-	-	No	No	No
2Ch/1Bh	Get Chassis Manager IPMB Address	§10.1.3.21	NA	NA	NA	No	No	No
2Ch/1Ch	Set Fan Policy	§10.1.3.15	NA	NA	NA	No	No	No
2Ch/1Dh	Get Fan Policy	§10.1.3.16	NA	NA	NA	No	No	No
2Ch/1Eh	FRU Control Capabilities	§10.1.3.18	O/M[3]	M	M	No	Yes	Yes
2Ch/1Fh	FRU Inventory Device Lock Control	§10.1.3.19	NA	NA	NA	No	No	No
2Ch/20h	FRU Inventory Device Write	§10.1.3.20	NA	NA	NA	No	No	No
2Ch/21h	Get Chassis Manager IP Addresses	§10.1.3.22	NA	NA	NA	No	No	No
2Ch/22h-3Fh	(reserved)	Reserved	-	-	-	No	No	No
2Ch/40h	Get FRU Address Info	§10.1.3.3	M	M	M	Yes	Yes	Yes
2Ch/41h	Get FRU Persistent Control	§10.1.3.23	O	O	O	No	No	No

NetFn/Cmd	Command	Reference	Req T1	Req T2	Req T3	Impl T1	Impl T2	Impl T3
2Ch/42h	Set FRU Persistent Control	§10.1.3.24	O	O	O	No	No	No
2Ch/43h	FRU Persistent Control Capabilities	§10.1.3.25	O	O	O	No	No	No
2Ch/44h	Get Mandatory Sensor Numbers	§10.1.3.26	O	O/M[4]	M	No	No	Yes
2Ch/45h	Get FRU Hash	§10.1.3.31	O	O	M	No	No	Yes
2Ch/46h	Get Payload Mode Capabilities	§10.1.3.32	O	O	M	No	No	Yes
2Ch/47h	Set Payload Mode	§10.1.3.33	O	O	M	No	No	Yes
2Ch/48h	Get Write Protect Capabilities	§10.1.3.34	O	O	O	No	No	No
2Ch/49h	Get Write Protect Enables	§10.1.3.35	O	O	O	No	No	No
2Ch/4Ah	Set Write Protect Enables	§10.1.3.36	O	O	O	No	No	No
2Ch/4Bh	Get Power Supply Capabilities	§10.1.3.37	O	O	O/M[5]	No	No	No
2Ch/4Ch	Get Power Supply Status	§10.1.3.38	O	O	O/M[5]	No	No	No
2Ch/4Dh	Set Power Supply Status	§10.1.3.39	O	O	O/M[5]	No	No	No
2Ch/4Eh	Get Control Bits Capabilities	§10.1.3.40	O	O	O	No	No	No
2Ch/4Fh	Get Control Bits	§10.1.3.41	O	O	O	No	No	No
2Ch/50h	Set Control Bits	§10.1.3.42	O	O	O	No	No	No

Bridged Firewall Commands

NetFn/Cmd	Command	Reference	Req T1	Req T2	Req T3	Impl T1	Impl T2	Impl T3
2Ch/51h	Get Bridged NetFn Support	§10.1.3.47	O	O	O	No	No	No
2Ch/52h	Get Bridged Command Enables	§10.1.3.43	O	O	O	No	No	No
2Ch/53h	Set Bridged Command Enables	§10.1.3.44	O	O	O	No	No	No
2Ch/54h	Get Bridged Command Sub-function Enables	§10.1.3.45	O	O	O	No	No	No
2Ch/55h	Set Bridged Command Sub-function Enables	§10.1.3.46	O	O	O	No	No	No
2Ch/56h	Set Bridged NetFn Policy	§10.1.3.48	O	O	O	No	No	No
2Ch/57h	Get Bridged NetFn Policy	§10.1.3.49	O	O	O	No	No	No

IPMI table requirement notes

- [2] Additional command-specific constraints apply in the IPMI definition.
- [3] The Device ID broadcast uses IPMB broadcast framing.
- [4] Chassis Manager operations apply to the whole chassis.
- [5] The Chassis Manager need not implement a full centralized SDR repository.
- [6] These commands are local Payload/System Interface operations.
- [7] Mandatory for a Tier-3 IPMC that includes a Payload Interface.
- [8] Mandatory for a Tier-3 IPMC with private I2C interfaces to subsidiary FRU EEPROMs.

VSO table requirement notes

- [2] Mandatory if Set FRU Activation is supported.
- [3] Mandatory if Set FRU Activation or FRU Control is supported.
- [4] Mandatory when the IPMC exposes more than one FRU.
- [5] Mandatory for an IPMC representing a power supply.

- [6] Mandatory for a Tier-3 IPMC representing a resettable payload.

Implementation scope

Platform Event is transmitted at Tier-2/3: this IPMC is an Event Generator, not an Event Receiver. Incoming Platform Event requests are not implemented. Chassis Manager functions, optional fan/LED control, subsidiary-FRU interfaces, IPMI firmware-firewall commands and VSO bridged-firewall commands are not claimed by the current implementation. Conditional M/O cells do not establish that an interface is present on this board. The complete source-table qualifications still apply.

Current applicability: one FRU (FRU 0), no IPMI Payload/System Interface, no private I2C subsidiary-FRU EEPROMs, and not a power-supply IPMC. The Tier-3 payload is resettable and the applicable FRU Control command is implemented. Accordingly:

- Set BMC Global Enables (06h/2Eh) and Send Message (06h/34h) are not implemented; their Tier-3 mandatory condition requires a Payload Interface.
- Master Write-Read (06h/52h) is not implemented; its Tier-3 mandatory condition requires a private subsidiary-FRU EEPROM interface.
- Get Power Supply Capabilities/Status and Set Power Supply Status (2Ch/4Bh-4Dh) are not implemented; the mandatory condition applies to power-supply IPMCs.
- Set/Get FRU State Policy Bits (2Ch/0Ah, 0Bh) are absent at Tier-1/2, where the mandatory condition requires Set FRU Activation, which those builds do not offer. All three activation/policy commands are implemented at Tier-3.
- FRU Control Capabilities (2Ch/1Eh) is absent at Tier-1, which implements neither FRU Control nor Set FRU Activation. It is implemented at Tier-2 and Tier-3.
- Get Mandatory Sensor Numbers (2Ch/44h) is absent at Tier-2, where its mandatory condition requires multiple FRUs. It is implemented at Tier-3.

Changing the hardware or interface scope can make an unimplemented O/M entry mandatory; review these conditions before offering that configuration.

The public Tier-1 software also supports firmware update over the console CLI and application UART through its bootloader. This is a public software feature, not an additional standardized IPMI/VSO firmware-update command. Runtime CLI entry is `DEBUG BOOTLOADER` (with `CONFIRM` when payload-power interruption requires it). The bootloader supports `INFO`, `ERASE APP`, `LOAD`, `VERIFY` and `RUN`; the public `tools/fwupdate/fwupdate.py` automates entry, erase, HEX loading, verification and restart. Ordinary APP update preserves the bootloader and stored FRU data.